Unit 7

Association Rule Mining

Association rules are if/then statements that help uncover relationships between seemingly unrelated data in a relational database or other information repository. An example of an association rule would be "*If a customer buys a dozen eggs, he is 80% likely to also purchase milk.*" An association rule has two parts, an antecedent (if) and a consequent (then). An antecedent is an item found in the data. A consequent is an item that is found in combination with the antecedent.

Association rule mining is a method for discovering interesting relations between variables in large databases. It is intended to identify strong rules discovered in databases using different measures of interestingness. For example, the rule found in the sales data of a supermarket would indicate that if a customer buys onions and potatoes together, they are likely to also buy hamburger meat. Such information can be used as the basis for decisions about marketing. In addition to the above example from market basket analysis association rules are employed today in many application areas including web usage mining, intrusion detection, bioinformatics etc.

The problem of association rule mining is defined as: Let $I = \{I1, I2.....In\}$ be a set of binary attributes called *items*. Let $D = \{T1, T2.....Tm\}$ be a set of transactions called the *database*. Each *transaction* in has a unique transaction ID and contains a subset of the items in A *rule* is defined as an implication of the form:

$$X => Y$$

Where $X, Y \subseteq I$ and $X \cap Y = \Phi$

Every rule is composed by two different set of items, also known as X and Y *itemsets* and, where X is called *antecedent* or left-hand-side (LHS) and Y is *consequent* or right-hand-side (RHS).

Transaction ID	Milk	Bread	Butter	Beer	Diaper
1	1	1	0	0	0
2	0	0	1	0	0
3	0	0	0	1	1
4	1	1	1	0	0
5	0	1	0	0	0

To illustrate the concepts, we use a small example from the supermarket domain. The set of items in above table shows a small database containing the items, where, in each entry, the value 1 means the presence of the item in the corresponding transaction, and the value 0 represent the absence of an item in a that transaction. An example rule for the supermarket could be $\{butter, bread\} => \{milk)$ meaning that if butter and bread are bought, customers also buy milk.

Support and Confidence

In order to select interesting rules from the set of all possible rules, constraints on various measures of significance and interest are used. The best-known constraints are minimum thresholds on *support and confidence*.

Support of association rule $A => B$ is the percentage of transactions in *dataset* that contain both items. In formula

$$Support(A => B) = P(A \cup B)$$

For example, in above data-set, the association rule $bread => milk$ has a support of *2/5* since both items occurs in *40%* of all transactions (2 out of 5 transactions).

Confidence of association rule $A => B$ with respect to set of transactions *T* in *dataset D is* the percentage of transactions in D containing *A* that also contain *B*. In formula

$$Confidence(A => B) = P(B \mid A) = P(A \cup B) / P(A) = Support(A => B) / P(A)$$

For example, in above data-set, the association rule $bread => milk$ has a confidence of *2/3,* since *66.66%* of all transactions containing *bread* also contains *milk*.

Rules that satisfy both a minimum support threshold and a minimum confidence threshold are called strong. By convention, we write support and confidence values so as to occur between 0% and 100%, rather than 0 to 1.0.

Why Association Mining

In data mining, association rules are useful for analyzing and predicting customer behavior. They play an important part in shopping basket data analysis, product clustering, and catalog design and store layout.
Programmers use association rules to build programs capable of machine learning

Apriori Algorithm

It is a classic algorithm used in data mining for learning association rules. It is very simple. Learning association rules basically means finding the items that are purchased together more frequently than others. The name of the algorithm is based on the fact that the algorithm uses *prior knowledge* of frequent item set properties.

Apriori employs an iterative approach known as a *level-wise* search, where *k*-itemsets are used to explore (*k*+1)-itemsets. First, the set of frequent 1-itemsets is found by scanning the database to accumulate the count for each item, and collecting those items that satisfy minimum support. The resulting set is denoted *L*1.Next, *L*1 is used to find *L*2, the set of frequent 2-itemsets, which is used to find *L*3, and so on, until no more frequent *k*-itemsets can be found. The finding of each *Lk* requires one full scan of the database. To improve the efficiency of the level-wise generation of frequent itemsets, an important property called the Apriori property, presented below, is used to reduce the search space. *Apriori Property states that any subset of frequent item set must be frequent*.

Example

TID	*List of item_IDs*
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

Consider a database, D, consisting of 9 transactions. Suppose min. support count required is 2 (i.e. min-sup = 2/9 = 22 %). Let minimum confidence required is 70%. We have to first find out the frequent item set using Apriori algorithm. Then, Association rules will be generated using min. support & min. confidence.

Solution

Step 1: Generating 1-itemset Frequent Pattern

The set of frequent 1-itemsets, L1, consists of the candidate 1-itemsets satisfying minimum support. In the first iteration of the algorithm, each item is a member of the set of candidate.

Scan D for count of each candidate →

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

C_1

Compare candidate support count with minimum support count →

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_1

In the first iteration of the algorithm, each item is a member of the set of candidate.

Step 2: Generating 2-itemset Frequent Pattern

To discover the set of frequent 2-itemsets, L2, the algorithm uses L1 *Join* L1to generate a candidate set of 2-itemsets, C2. Next, the transactions in D are scanned and the support count for each candidate item set in C2 is accumulated (as shown in the middle table). The set of frequent 2-itemsets, L2, is then determined, consisting of those candidate 2-itemsets in C2 having minimum support.

Generate C_2 candidates from L_1 →

Itemset
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

C_2

Scan D for count of each candidate →

Itemset	Sup. Count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

C_2

Compare candidate support count with minimum support count →

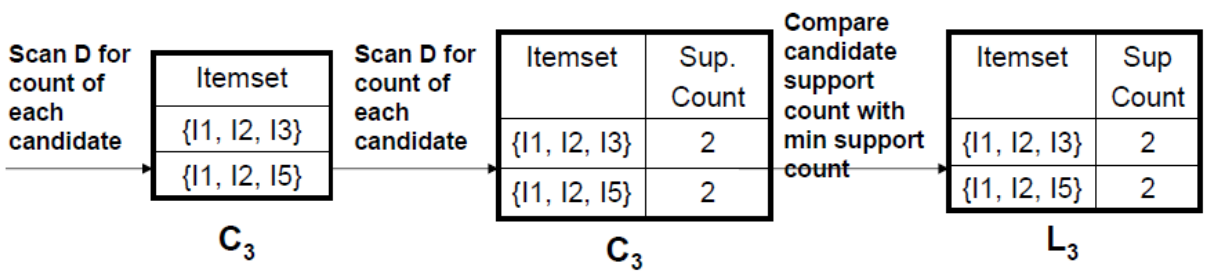
Itemset	Sup Count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L_2

Step 3: Generating 3-itemset Frequent Pattern

The generation of the set of candidate 3-itemsets, C3, involves use of the Apriori Property. In order to find C3, we compute L2*Join*L2. C3= L2 *Join*L2 = {{I1, I2, I3}, {I1, I2, I5}, {I1, I3, I5}, {I2, I3, I4}, {I2, I3, I5}, {I2, I4, I5}}. Now, Join step is complete and Prune step will be used to reduce the size of C3. Prune step helps to avoid heavy computation due to large Ck.

Based on the Apriori property that all subsets of a frequent item set must also be frequent, we can determine that four latter candidates cannot possibly be frequent. For example, let's take {I1, I2, I3}.The 2-item subsets of it are {I1, I2}, {I1, I3} & {I2, I3}. Since all 2-item subsets of {I1, I2, I3} are members of L2, We will keep {I1, I2, I3} in C3. Lets take another example of {I2, I3, I5} which shows how the pruning is performed. The 2-item subsets are {I2, I3}, {I2, I5} & {I3,I5}. BUT, {I3, I5} is not a member of L2and hence it is not frequent violating Apriori Property. Thus we will have to remove {I2, I3, I5} from C3. Therefore, C3= {{I1, I2, I3}, {I1, I2, I5}} after checking for all members of result of Join operation for Pruning. Now, the transactions in D are scanned in order to determine L3, consisting of those candidates 3-itemsets in C3 having minimum support.



Step 4: Generating 4-itemset Frequent Pattern

The algorithm uses L3 *Join*L3to generate a candidate set of 4-itemsets, C4. Although the join results in {{I1, I2, I3, I5}}, this item set is pruned since its subset {{I2, I3, I5}}is not frequent. Thus, C4= φ, and algorithm terminates, having found all of the frequent items. This completes our Apriori Algorithm.

These frequent itemsets will be used to generate strong association rules (where strong association rules satisfy both minimum support & minimum confidence).

Step 5: Generating Association Rules from Frequent Itemsets

Procedure:

For each frequent item set ***"l"***, generate all nonempty subsets of ***l.*** For every nonempty subset ***s*** of ***l***, output the rule "$s => l - s$"if support_count(*l*) / support_count(s) >= **min_conf** where min_conf is minimum confidence threshold.

Back To Example

We had L = {{I1}, {I2}, {I3}, {I4}, {I5}, {I1,I2}, {I1,I3}, {I1,I5}, {I2,I3}, {I2,I4}, {I2,I5}, {I1,I2,I3}, {I1,I2,I5}}.

Let's take ***l*** = {I1, I2, I5}. It's all nonempty subsets are {I1, I2}, {I1, I5}, {I2, I5}, {I1}, {I2}, {I5}.

Let minimum confidence threshold is, say 70%. The resulting association rules are shown below, each listed with its confidence.

R1: I1 ^ I2 =>I5

Confidence = support_count {I1, I2,

I5}/support_count{I1,I2} = 2/4 = 50%, R1 is Rejected.

R2: I1 ^ I5 =>I2

Confidence = support_count {I1,I2,I5}/ support_count {I1,I5} = 2/2 = 100%, R2 is Selected.

R3: I2 ^ I5 =>I1

Confidence = support_count {I1,I2,I5}/ support_count {I2,I5} = 2/2 = 100%, R3 is Selected.

R4: I1 =>I2 ^ I5

Confidence = support_count {I1,I2,I5}/ support_count {I1} = 2/6 = 33%, R4 is Rejected.

R5: I2 =>I1 ^ I5

Confidence = support_count {I1,I2,I5}/ support_count {I2} = 2/7 = 29%, R5 is Rejected.

R6: I5 =>I1 ^ I2

Confidence = support_count {I1,I2,I5}/ support_count {I5} = 2/2 = 100%, R6 is Selected.

In this way, we have found three strong association rules.

Improving Efficiency of Apriori Algorithm

Many variations of the Apriori algorithm have been proposed that focus on improving the efficiency of the original algorithm. Several of these variations are summarized as follows:

Hash-based technique

A hash-based technique can be used to reduce the size of the candidate *k*-itemsets, *Ck*, for $k > 1$. For example, when scanning each transaction in the database to generate the frequent 1-itemsets, *L*1, from the candidate 1-itemsets in *C*1, we can generate all of the 2-itemsets for each transaction, hash them into the different *buckets* of a *hash table* structure, and increase the corresponding bucket counts. A 2-itemset whose corresponding bucket count in the hash table is below the support threshold cannot be frequent and thus should be removed from the candidate set. Such a hash-based technique may substantially reduce the number of the candidate *k*-itemsets examined.

Create hash table H_2 using hash function $h(x, y) = ((order\ of\ x) \times 10 + (order\ of\ y))\ mod\ 7$ →

H_2

bucket address	0	1	2	3	4	5	6
bucket count	2	2	4	2	2	4	4
bucket contents	{I1, I4} {I3, I5}	{I1, I5} {I1, I5}	{I2, I3} {I2, I3} {I2, I3} {I2, I3}	{I2, I4} {I2, I4}	{I2, I5} {I2, I5}	{I1, I2} {I1, I2} {I1, I2} {I1, I2}	{I1, I3} {I1, I3} {I1, I3} {I1, I3}

Transaction Reduction

It reduces the number of transactions scanned in future iterations. A transaction that does not contain any frequent *k*-itemsets cannot contain any frequent (*k*+1)-itemsets. Therefore, such a transaction can be marked or removed from further consideration because subsequent scans of the database for *j*-itemsets, where $j > k$, will not require it.

Partitioning

The set of transactions may be divided into a number of disjoint subsets. Then each partition is searched for frequent itemsets. These frequent itemsets are called local frequent itemsets. *Any itemset that is potentially frequent with respect to D must occur as a frequent itemset in at least one of the partitions*. Therefore, all local frequent itemsets are candidate itemsets with respect to *D*. The collection of frequent itemsets from all partitions forms the global candidate itemsets with respect to *D*.

Sampling

A random sample (usually large enough to fit in the main memory) may be obtained from the overall set of transactions and the sample is searched for frequent itemsets. These frequent itemsets are called *sample frequent itemsets*. Because we are searching for frequent itemsets in *S* rather than in *D*, it is possible that we will miss some of the global

frequent itemsets. To lessen this possibility, we use a lower support threshold than minimum support to find the frequent itemsets local to S